BulSUScholar — Exact Thesis Revision and Replacement Guide

A manual, copy-and-paste editing roadmap for A Centralized Scholarship Management and Tracking System for Bulacan State University – Bustos Campus. The original manuscript has NOT been edited.

Every entry identifies the thesis section, an existing search phrase, whether to REPLACE/ADD/KEEP text, and the precise proposed wording. Verify all implementation claims and reported tests against the current deployed code and evidence before submission. Printed pages can shift after revision.

CHAPTER I — THE PROBLEM AND ITS BACKGROUND

R1-A. Scope of the Study

LOCATION: Chapter I > Scope and Limitations > Scope of the Study; approximately page 9

FIND IN WORD: The system manages student scholarship records and supports pre-approval validation

ACTION: REPLACE the entire paragraph beginning with that phrase; KEEP the following paragraph beginning “The system also provides scholarship recommendations”.

WHAT TO PUT INSTEAD:

The system centralizes student accounts, scholarship applications, uploaded documentary requirements, internal scholarship records, imported official scholarship rosters, scholarship commitments, monitoring information, and selected audit records within a unified database. The system supports student-record verification and scholarship validation to help prevent conflicting scholarship assignments.

External scholarship providers, including local government units and other organizations, may be represented by grantor accounts classified as External. Authorized users can upload official scholarship recipient lists through the existing spreadsheet roster-import process. Government, Private, and Others remain separate provider classifications. External scholarship records become available for identity matching and scholarship-assignment validation after they have been entered and validated within the system.

R1-B. Conceptual Framework — process explanation

LOCATION: Chapter I > Conceptual Framework; approximately page 8

FIND IN WORD: After application submission, the system performs pre-approval validation

ACTION: REPLACE the full paragraph beginning with this sentence; KEEP the next paragraph beginning “The output stage represents”.

WHAT TO PUT INSTEAD:

After application submission, the system performs validation at different stages of the scholarship management process. These include student registration verification, document validation, official scholarship roster-import checks, and scholarship commitment validation. The system uses available internal and validated external scholarship records to identify possible duplicate identities and conflicting scholarship assignments.

Imported scholarship rosters may also be reconciled when a matching student account is activated or completes verified login. Possible identity conflicts are referred to a Full Administrator for verification. Account activation, document review, scholarship application approval, and official scholarship assignment are treated as separate operations.

Once the applicable validation and approval procedures are completed, the system updates the corresponding scholarship records and supports subsequent monitoring, renewal, notification, and reporting activities.

R1-C. Limitations of the Study

LOCATION: Chapter I > Limitations of the Study; approximately page 10

FIND IN WORD: The system does not provide full real-time system-to-system integration

ACTION: REPLACE that entire paragraph; KEEP the following paragraph beginning “The accuracy of the system depends”.

WHAT TO PUT INSTEAD:

The system does not provide full real-time system-to-system integration with external scholarship providers such as DOST, UNIFAST, local government scholarship offices, and other government or private scholarship organizations. Externally sourced scholarship information must first be uploaded through the official roster-import process, manually encoded where supported, or maintained by authorized users.

The system can compare only scholarship information that has been entered and sufficiently updated within its centralized database. Missing, incomplete, or outdated external scholarship records may therefore limit the identification of conflicting scholarship assignments. Direct integration with external scholarship databases remains a possible future enhancement, subject to institutional authorization, data-sharing agreements, and appropriate security requirements.

CHAPTER II — REVIEW OF RELATED LITERATURE AND STUDIES

R2. Explain the rationale for Record Linkage and Levenshtein

LOCATION: Chapter II > Technical Background > Duplication Prevention

FIND IN WORD: The duplicate checker applies different matching rules depending on the available information

ACTION: ADD AFTER the paragraph explaining 72%, 82%, and 84% thresholds, BEFORE the Technology Acceptance Model heading. KEEP the weights, thresholds, and references.

WHAT TO PUT INSTEAD:

The use of weighted Record Linkage with Levenshtein-based similarity provides an interpretable approach for identifying potentially duplicate student records that may contain minor spelling differences, typographical errors, or formatting inconsistencies. Data normalization reduces differences caused by capitalization, punctuation, and spacing, while Levenshtein similarity measures variations between text values based on character insertions, deletions, and substitutions.

The application of attribute weights allows the comparison process to consider multiple student identifiers rather than relying solely on the student's name. This is important because different students may possess similar or identical names despite having different personal and academic records.

However, a high similarity score does not automatically establish that two records belong to the same individual. Consequently, similarity-based results are treated as indicators of possible duplicate identities. Official scholarship assignment requires separate verification of student identity, the applicable academic cycle, recognized scholarship information, and existing scholarship commitments. This distinction reduces the risk of treating a possible duplicate match as sufficient authorization for a scholarship award.

CHAPTER III — METHODOLOGY AND SYSTEM MODULES

R2-B. Actual duplication-prevention decision rules

LOCATION: Chapter III > System Modules > 6. Duplication Prevention Algorithm

FIND IN WORD: Separate validation rules are also used in other workflows

ACTION: REPLACE this paragraph only; KEEP the formulas and weight table.

WHAT TO PUT INSTEAD:

The system implements different validation procedures according to the workflow being performed. The general duplicate checker compares Student ID, full name, email address, contact number, course, year level, city or municipality, and province using exact matching and weighted similarity calculations. Available comparison values contribute to the normalized weighted similarity score.

An exact normalized Student ID may independently identify a possible duplicate. An exact email address or contact number combined with at least 72% name similarity may also identify a possible duplicate. The general fuzzy-matching condition requires at least 82% name similarity and 84% weighted similarity. These configured values represent matching thresholds rather than experimentally measured accuracy percentages.

Official scholarship roster assignment uses separate validation rules. An automatic assignment requires an exact Student ID match, a matching normalized full name, a recognized scholarship program, a verified academic cycle, and the absence of a conflicting current-cycle award.

Therefore, a fuzzy-name match alone does not authorize a scholarship assignment. Potential identity matches may be flagged for review, while confirmed scholarship conflicts are handled through the applicable validation and administrative resolution process.

R3-A. Approval Module

LOCATION: Chapter III > System Modules > 7. Approval Module

FIND IN WORD: The approval module provides scholarship grantors and coordinators

ACTION: REPLACE the paragraph under the Approval Module heading.

WHAT TO PUT INSTEAD:

The Approval Module supports the management of scholarship application decisions and official scholarship assignments while maintaining separate procedures for student account activation, academic-document validation, and Student Application Profile review.

During registration, student identity information and the applicable signup requirements are validated. When Manual Account Review is enabled, a Full Administrator reviews the required registration documents and approves the account after email confirmation and the necessary validation checks. When Manual Account Review is disabled, eligible accounts may activate automatically after satisfying the applicable requirements.

Certificates of Registration (COR) and applicable Reports of Grades (ROG) undergo automatic parsing and policy validation. Valid submissions are accepted according to the implemented rules, while invalid submissions require correction. Student Application Profile submissions remain subject to administrator review. Student identification is verified during account creation, while optional Student ID replacements by active students may be processed according to the implemented automatic validation rules.

Scholarship grantors and coordinators review applications and supporting information before performing authorized scholarship decisions. Once an official scholarship assignment or applicable commitment is confirmed, the system

preserves the existing award, archives competing current-cycle pending applications where required, and releases their reserved slots without deleting historical records or submitted documents.

These procedures ensure that account activation, document acceptance, application approval, and scholarship assignment are not treated as one universal approval operation.

R3-B. Figure 3.1 registration/authentication

LOCATION: Chapter III > Explanation below Figure 3.1

FIND IN WORD: Figure 3.1 presents the student account registration and authentication process

ACTION: REPLACE the explanation below Figure 3.1; UPDATE the actual figure.

WHAT TO PUT INSTEAD:

Figure 3.1 presents the student account registration and authentication process. Students provide the required registration information and identity documents, which undergo the applicable system validation checks. Following email confirmation, the system determines whether Manual Account Review is enabled.

When Manual Account Review is enabled, a Full Administrator reviews the applicable signup documents and approves the account after the required checks have been satisfied. Otherwise, the account may be activated automatically if all applicable requirements are met.

After activation, the system may reconcile eligible previously imported official scholarship rosters using verified student identity information. Any identity mismatch or conflicting scholarship assignment is referred to the authorized administrative resolution process rather than automatically granting another scholarship. Account activation and scholarship assignment remain separate system operations.

Figure edit: Email verified? > Signup requirements passed? > Manual Account Review enabled? > Account activation > Roster reconciliation > Identity/award conflict? Route conflicts to Full Admin.

R3-C. Figure 3.2 profile/document verification

LOCATION: Chapter III > Explanation below Figure 3.2

FIND IN WORD: Figure 3.2 presents the process for completing and verifying

ACTION: REPLACE the explanation; UPDATE diagram with separate automatic COR/ROG and administrator-profile branches.

WHAT TO PUT INSTEAD:

Figure 3.2 illustrates the process of completing student profile information and validating documentary requirements. Certificates of Registration (COR) and applicable Reports of Grades (ROG) are processed through automatic document parsing and policy validation. When the submitted document satisfies the applicable requirements, the system accepts it. Otherwise, the student receives a correction message or the corresponding validation result.

Student Application Profile submissions follow a separate procedure and remain pending until reviewed by an authorized administrator. Student identification is required during registration but is not treated as a recurring semester-document requirement. Active students may replace their previously submitted identification documents where the system permits.

The Manual Account Review setting applies to new-account activation and does not convert routine COR and ROG validation into a universal manual document-review process.

R4. Figure 3.5 official roster import

LOCATION: Chapter III > System Process > Explanation below Figure 3.5

FIND IN WORD: Figure 3.5 presents the process for importing an official scholar roster

ACTION: REPLACE entire explanation; UPDATE Figure 3.5 with preview, identity, academic-cycle, conflict, active-account, and resolution decisions.

WHAT TO PUT INSTEAD:

Figure 3.5 illustrates the process of importing official scholarship recipient rosters and validating scholarship assignments. Authorized users upload a spreadsheet containing official scholarship recipient information through the roster-import module. The process supports both Internal and External grantors, with External representing a provider classification rather than a separate administrative authorization role.

During the preview stage, the system checks required student identity information, repeated entries, recognized scholarship programs, academic-cycle information, and possible scholarship-assignment conflicts. Before the import is committed, the system repeats the applicable validation checks within a database transaction to prevent inconsistent assignments.

For an active student whose Student ID and normalized full name match the official roster, the system may confirm a valid scholarship assignment for the applicable academic cycle. An existing matching pending application may be converted into the confirmed assignment without replacing its application identifier or deleting its submitted documents and progress records. Competing pending applications within the current cycle are archived, and their reserved slots are released.

If a matching student account is not yet active, the validated roster record is retained for reconciliation during account activation or subsequent verified login. Repeated roster processing does not create duplicate scholarship assignments or duplicate notifications.

When the system detects a mismatched identity, an unverified academic cycle, multiple competing roster alternatives, or an existing conflicting scholarship award, automatic assignment is prevented. The conflict is displayed for authorized review through the corresponding warning interface. A Full Administrator verifies the supporting information and records the required reason before resolving the conflict. Existing awards and historical records are preserved.

R5. Monitoring Module — academic-cycle logic

LOCATION: Chapter III > 8. Monitoring Module

FIND IN WORD: This module enabled continuous tracking of scholarship records

ACTION: KEEP existing monitoring paragraph. ADD AFTER it, BEFORE 9. Notification Module.

WHAT TO PUT INSTEAD:

Scholarship validation and monitoring are evaluated according to the applicable academic cycle and the authoritative application and scholarship commitment records. The academic cycle refers to the academic year and semester associated with a scholarship record.

A scholarship award that has been confirmed within the current academic cycle remains recognized as an existing commitment even when its processing workflow has been marked as Finished or Completed. Therefore, completion of the scholarship workflow does not automatically permit another conflicting scholarship assignment during the same academic cycle.

Completed awards from explicitly recorded previous academic cycles are maintained as historical records and do not independently prevent eligible applications for the current cycle. Cancelled, terminated, and inactive scholarship

commitments are not treated as active-award blockers, although other scholarship eligibility requirements continue to apply.

Pending applications that have not resulted in a confirmed scholarship award or commitment are distinguished from active scholarship commitments. If an official scholarship assignment is confirmed, competing pending applications for the applicable academic cycle may be archived while retaining their associated records and uploaded documents.

Legacy scholarship records without verified academic-cycle information require authorized administrative review before assignment. The system does not automatically assume the academic year or semester of an incomplete historical record.

ADD a six-row status table after paragraph (current awarded; current completed; previous completed; cancelled/terminated; pending; legacy without cycle). In Approval Module replace “removed or disabled from other applicant lists” with “Competing current-cycle pending applications are archived, and their reserved slots are released when the official assignment or applicable commitment transaction succeeds. Existing application documents and historical scholarship records are preserved.”

R6. Targeted testing methodology

LOCATION: Chapter III > Testing

FIND IN WORD: Functional testing verified whether major system modules performed their expected functions

ACTION: KEEP existing functional testing paragraph; ADD AFTER it and BEFORE the TAM/ISO assessment paragraph.

WHAT TO PUT INSTEAD:

In addition to the selected core functional tests, targeted automated testing was conducted to verify student-record matching, official scholarship roster import, identity reconciliation, scholarship assignment, and prevention of conflicting current-cycle awards.

Backend regression tests examined exact identity matching, configured similarity thresholds, account activation and verified-login requirements, role-based authorization, external grantor classification, and scholarship conflict handling.

Focused database tests were performed using isolated PostgreSQL/PGlite environments and synthetic student and scholarship records. These tests assessed roster assignment, repeated processing, current-cycle award conflicts, previous-cycle historical records, cancelled or terminated commitments, competing pending applications, reservation release, administrative restrictions, and transaction rollback behavior.

Each test was documented using an identifiable test-case number, a description of the function being evaluated, expected results, actual observed results, and the corresponding test status.

These automated technical tests were separate from the system acceptability assessment using TAM and the software quality evaluation using ISO/IEC 25010. Successful technical test execution does not independently establish the accuracy of document extraction from actual student documents or the completeness of external scholarship records.

R6-B. Table 1 testing discussion

LOCATION: Chapter III > Testing > text accompanying Table 1

FIND IN WORD: Table 1 presents the summary of the selected core test cases

ACTION: REPLACE Table 1 discussion paragraph; KEEP original 15 selected cases if corroborated. VERIFY numerical counts before pasting.

WHAT TO PUT INSTEAD:

Table 1 presents 15 selected core automated backend test cases conducted to verify the functionality of major system modules. The table summarizes the tested operations, their expected outcomes, actual observed results, and corresponding test status.

Additional targeted verification activities were recorded separately from the selected core test cases. The implementation verification documented on October 23, 2026, reported 348 passing backend tests, including 23 added Solis-specific regression tests. A separate focused database testing process reported 27 passing synthetic test cases. Seven frontend security checks, frontend lint verification, and the production build were also reported as successful.

These results represent different sets of technical verification activities and were not combined into a single algorithm accuracy percentage or treated as equivalent to real-user evaluation. The results are limited to the documented testing environments and datasets.

CHAPTER IV — RESULTS AND DISCUSSION

R7-A. Specific Objective No. 2 results subsection

LOCATION: Chapter IV > II. Designing and Developing ... > Validation Module discussion

FIND IN WORD: The Validation Module supports student-record verification

ACTION: KEEP the existing paragraph. ADD AFTER it, BEFORE “The Approval Module allows authorized scholarship grantors...”. Add subheading “Results for Specific Objective No. 2: Record Validation and Prevention of Multiple Active Scholarships”.

WHAT TO PUT INSTEAD:

The developed system's student-record verification and duplicate-scholarship prevention functions were evaluated through controlled test cases involving student identity information, internal scholarship records, and official scholarship roster records available within the system.

The tests examined exact identity matching, possible duplicate-record identification, academic-cycle validation, existing scholarship commitment checks, roster reconciliation, and administrative conflict resolution.

The focused database verification demonstrated that existing current-cycle scholarship awards were preserved when competing assignments were attempted. Completed scholarship records from previous academic cycles were maintained as historical records and did not independently prevent current-cycle eligibility processing.

Additional tests evaluated mismatched student identities, repeated roster-import operations, competing pending applications, administrator authorization restrictions, and the preservation of submitted documents and scholarship records.

The reported results support the expected behavior of the tested implementation under controlled synthetic conditions. However, these results do not establish complete detection of scholarships from unrecorded external providers, measured extraction accuracy using actual COR and ROG documents, or successful deployment of the revised implementation in the live production environment.

ADD after this paragraph a selected SOL-SYS table using verified case results. Suggested IDs: 02, 04, 05, 06, 11, 17. Specify isolated/synthetic environment.

R7-B. Figure 4.11 duplicate scholarship validation

LOCATION: Chapter IV > Explanation below Figure 4.11

FIND IN WORD: Figure 4.11 presents the duplicate scholarship validation feature

ACTION: REPLACE explanation under Figure 4.11; STOP before Figure 4.12 Approval Page.

WHAT TO PUT INSTEAD:

Figure 4.11 presents the duplicate-scholarship validation feature of the developed system. The validation process checks available student and scholarship records to identify potential duplicate identities and conflicting scholarship commitments.

General duplicate-record checking uses exact identity comparisons and similarity-based matching to identify records that may refer to the same individual. Possible duplicate identities may be flagged for authorized review. However, a similarity-based warning alone does not establish that a student has already received a scholarship or authorize another scholarship assignment.

Official scholarship roster assignment uses separate checks involving student identity, recognized scholarship information, academic-cycle verification, and existing scholarship commitments. A confirmed competing scholarship award within the current academic cycle prevents an additional conflicting application or automatic assignment.

Conflicts requiring administrative intervention are referred to an authorized Full Administrator through the corresponding warning interface, with documented resolution reasons. This process supports the preservation of existing scholarship records and helps prevent conflicting scholarship assignments using information available within the system.

R8-A. Document extraction testing METHOD

LOCATION: Chapter III > Testing (place after R6 technical testing, before TAM/ISO)

FIND IN WORD: In this phase, the developed system underwent testing procedures

ACTION: ADD subheading “Document Extraction Accuracy Evaluation” and following method. Use completed-test past tense only if performed.

WHAT TO PUT INSTEAD:

The document-extraction accuracy evaluation assesses the system's ability to identify and validate student information from Certificates of Registration (COR) and applicable Reports of Grades (ROG).

The evaluation compares the information extracted by the document-processing mechanism against manually verified reference values obtained from authorized document samples. The assessment considers the correct identification of Student IDs, applicable academic information, and supported grade-related fields.

The evaluation includes text-based PDF documents, scanned PDF documents, low-quality copies, documents containing easily confused characters such as 7 and 1, and documents with missing or incorrect information.

The results are classified as correctly extracted information, incorrectly extracted information, and rejected or unreadable documents. Rejected documents are documented separately rather than automatically classified as successful extraction.

This evaluation is conducted separately from the general duplicate-record matching tests and official scholarship-assignment validation. Its purpose is to identify the accuracy and limitations of the document-processing mechanism under the evaluated conditions.

R8-B. Document accuracy RESULTS — template only

LOCATION: Chapter IV > Objective No. 2 results subsection

FIND IN WORD: New selected SOL-SYS test results table from R7

ACTION: ADD AFTER selected test results table. DO NOT claim completion until tests are actually run.

WHAT TO PUT INSTEAD:

The document-extraction evaluation was conducted using [number] authorized COR and ROG samples representing different document formats and quality conditions.

The extracted student information was compared against manually verified reference values to determine the correctness of Student ID recognition and other supported document fields.

Among the evaluated samples, [number] produced the expected extracted information, [number] contained extraction discrepancies, and [number] were rejected or could not be processed.

The evaluation identified [describe actual findings], particularly in relation to [describe observed issues, if any]. These results provide evidence of the document-processing mechanism's performance under the evaluated conditions and identify areas requiring further improvement.

ADD sample table DOC-001 text PDF, DOC-002 clear scan, DOC-003 low-quality/7-vs-1, DOC-004 wrong-owner/cycle; record actual extracted/reference values and outcomes. Do not disclose unmasked student data.

CHAPTER V — SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

R9-A. Summary of Findings No. 2

LOCATION: Chapter V > Summary of Findings

FIND IN WORD: 2. The system implements a pre-approval validation mechanism

ACTION: REPLACE Finding No. 2 only.

WHAT TO PUT INSTEAD:

2. The developed system implements student-record verification, duplicate-record detection, and academic-cycle-aware scholarship-assignment safeguards using available internal scholarship information and validated external scholarship rosters stored within the centralized database. The validation mechanism employs exact identity checks, weighted similarity comparisons, and scholarship commitment validation to identify potential duplicates and prevent conflicting awards within the applicable academic cycle. Controlled technical tests support the implemented validation behavior under the evaluated conditions. However, the completeness of external scholarship detection remains dependent on the availability and accuracy of records entered into the system.

R9-B. Conclusions

LOCATION: Chapter V > Conclusions

FIND IN WORD: The pre-approval validation mechanism supports the prevention of duplicate scholarship allocation

ACTION: REPLACE validation conclusion paragraph only.

WHAT TO PUT INSTEAD:

The implemented validation and scholarship-assignment mechanisms support the prevention of conflicting scholarship awards through student-record verification, duplicate-detection procedures, and academic-cycle-aware commitment checks. The system distinguishes possible duplicate identities from confirmed scholarship assignments and preserves existing current-cycle awards when conflicting assignments are attempted. Previous-cycle scholarship records are retained for monitoring and historical reference.

These mechanisms strengthen the management of scholarship records available within the centralized database. However, the system does not automatically identify scholarships that have not been entered or updated by authorized users, and similarity-based matching alone does not establish definitive student identity or scholarship-award authorization.

R9-C. Recommendation — student verification

LOCATION: Chapter V > Recommendations

FIND IN WORD: The researchers recommend improving student-record verification

ACTION: REPLACE that substantive recommendation paragraph.

WHAT TO PUT INSTEAD:

The researchers recommend further evaluating and refining the implemented student-record verification and duplicate-scholarship prevention mechanisms through additional testing using larger authorized datasets. Future improvements may focus on minimizing false-positive identity matches, strengthening the validation of incomplete student information, and improving the reliability of document extraction for low-quality or difficult-to-read submissions.

These improvements are intended to strengthen the existing validation capabilities rather than introduce duplicate prevention as an entirely new functionality.

R9-D. Recommendation — external integration

LOCATION: Chapter V > Recommendations

FIND IN WORD: The researchers also recommend the future integration of external scholarship databases

ACTION: REPLACE external integration recommendation paragraph.

WHAT TO PUT INSTEAD:

The researchers also recommend exploring future direct integration with authorized external scholarship databases and providers. Although the developed system already supports the inclusion of externally sourced scholarship information through official roster imports and authorized record management, direct system-to-system integration may improve the completeness, timeliness, and consistency of scholarship information.

Such integration would require appropriate institutional permissions, data-sharing agreements, security safeguards, and cooperation with participating scholarship providers. Future integration should complement the existing roster-import functionality while strengthening scholarship validation and monitoring capabilities.

ADDITIONAL REVISIONS REQUESTED BY THE RESEARCHERS

R10-A. Replace IPO framework opening (conditional)

LOCATION: Chapter I > Conceptual Framework

FIND IN WORD: This study is anchored on the Input–Process–Output (IPO) model

ACTION: REPLACE opening paragraph only IF adviser approves DSRM as the research framework.

WHAT TO PUT INSTEAD:

This study is guided by the Design Science Research Methodology (DSRM) proposed by Peffers et al. (2007), which provides a structured approach to developing and evaluating technological artifacts designed to address identified

organizational problems. The methodology consists of six major activities: problem identification and motivation, definition of the objectives of a solution, design and development, demonstration, evaluation, and communication. These activities provide a research-oriented structure for explaining the development and evaluation of the centralized scholarship management and tracking system for Bulacan State University – Bustos Campus.

R10-B. Figure 1.1 — diagram

LOCATION: Chapter I > Figure 1.1

FIND IN WORD: Figure 1.1. Conceptual Framework of the Study

ACTION: REPLACE IPO graphic with six DSRM stage boxes IF approved. Retain caption number.

WHAT TO PUT INSTEAD:

1. Problem Identification — fragmented records, manual processes and duplicate-award concerns.
2. Objectives of the Solution — centralization, validation, recommendation and monitoring.
3. Design and Development — BulSUScholar application, database, algorithms and workflows.
4. Demonstration — registration, roster import, assignment, approval and tracking.
5. Evaluation — functional tests, technical checks, TAM and ISO/IEC 25010.
6. Communication — findings, conclusions and recommendations.

R10-C. Explain revised Figure 1.1

LOCATION: Chapter I > Conceptual Framework, below Figure 1.1

FIND IN WORD: Figure 1.1 presents the conceptual framework of the study using the Input–Process–Output

ACTION: REPLACE Figure 1.1 explanation and subsequent IPO-specific input/process/output explanations, STOP before Scope and Limitations. IF approved.

WHAT TO PUT INSTEAD:

Figure 1.1 presents the conceptual framework of the study based on the six major activities of the Design Science Research Methodology (DSRM).

The problem identification stage focuses on the challenges associated with fragmented scholarship records, manual application processing, difficulties in monitoring scholarship activities, and the possibility of conflicting scholarship assignments.

The objectives of the solution stage establishes the need for a centralized web-based system that supports scholarship posting, student applications, document submission, verification, recommendation, monitoring, and reporting.

The design and development stage includes the creation of the system architecture, database structure, user interfaces, scholarship management modules, rule-based recommendation mechanism, and duplicate-detection functionality.

The demonstration stage involves presenting the developed system's major functions and workflows to illustrate how scholarship-related activities are supported through the centralized platform.

The evaluation stage assesses the system using functional and technical testing, the Technology Acceptance Model (TAM), and the ISO/IEC 25010 software quality model.

Finally, the communication stage presents the research findings, system evaluation results, conclusions, and recommendations. Together, these activities explain the relationship between the identified problems, the developed technological solution, and its evaluation within the context of scholarship management at Bulacan State University –

Bustos Campus.

Research citation: Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. Journal of Management Information Systems, 24(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302. IMPORTANT: DSRM is a methodology, not a conventional theory; reconcile Chapter III Developmental Research Design and Agile references rather than silently substituting.

R11-A. Actual code snippet — scholarship recommendations

LOCATION: Chapter III > Rule-Based Decision Tree Scholarship Recommendation discussion

FIND IN WORD: Existing subsection explaining the rule-based Decision Tree

ACTION: ADD AFTER the existing decision-rule explanation; BEFORE next module. Add heading “Implementation of the Rule-Based Scholarship Recommendation Algorithm”.

WHAT TO PUT INSTEAD:

The scholarship recommendation algorithm evaluates student information against the predefined eligibility conditions established for each scholarship program. The system retrieves the applicable student attributes and scholarship requirements and evaluates the corresponding decision conditions to determine whether a scholarship may be recommended.

The algorithm does not require a historical training dataset because the recommendation process is based on predefined eligibility rules rather than a trained machine-learning model.

The following code excerpt illustrates the implementation of the scholarship recommendation decision process.

[Insert the actual relevant source-code excerpt here.]

The code demonstrates how the eligibility conditions are evaluated, how applicants are excluded from recommendations when mandatory conditions are not satisfied, and how qualifying scholarship opportunities are identified. The recommendation result serves as decision support and does not automatically constitute scholarship approval.

ADD a worked example afterward with a real scholarship rule set and sample student. Do not insert invented implementation code.

R11-B. Actual code snippets — Record Linkage / Levenshtein

LOCATION: Chapter III > 6. Duplication Prevention Algorithm

FIND IN WORD: When the system identifies a possible duplicate or conflict

ACTION: ADD AFTER this paragraph, BEFORE 7. Approval Module. Add heading “Implementation of the Record Linkage and Duplicate-Detection Algorithm”.

WHAT TO PUT INSTEAD:

The duplicate-record validation algorithm begins by normalizing the available student information to ensure that differences in capitalization, punctuation, and formatting do not unnecessarily affect the comparison process.

The system calculates field-level similarities using the applicable comparison methods and evaluates the results according to the configured attribute weights and matching thresholds.

The following implementation excerpts illustrate the data-normalization process, similarity calculation, weighted record comparison, and duplicate-detection conditions.

[Insert actual normalization function excerpt.]

[Insert actual Levenshtein similarity and weighted comparison excerpt.]

[Insert actual duplicate-matching conditions excerpt.]

The general duplicate-detection result is used to identify potentially matching student records. It is not sufficient, by itself, to authorize an official scholarship assignment.

Official roster assignment applies a separate set of validation requirements involving exact student identity, normalized full name, recognized scholarship information, academic-cycle verification, and existing scholarship commitments.

[Insert actual official roster-assignment validation excerpt.]

These excerpts demonstrate the distinction between identifying possible duplicate student records and preventing unauthorized or conflicting scholarship assignments.

Relevant sources named in revision guide: backend/grantor_algorithms.py and roster workflow services. Those files were NOT included with the thesis documents; replace placeholders with verified actual code, not illustrative code.

R12 — TABLES, PAGINATION, AND FINAL FORMATTING

• In each long table: select table > right-click > Table Properties > Row > UNCHECK “Allow row to break across pages”.

• For multi-page tables: select heading row > Table Layout > Repeat Header Rows.

• For tables extending past the right margin: Table Layout > AutoFit > AutoFit to Window. Adjust column widths and padding rather than shrinking text until unreadable.

• A table may legitimately span multiple pages. Avoid clipping and split individual rows when possible; repeat column headers.

• Update the Table of Contents only AFTER all content edits: References > Update Table > Update entire table. Export to PDF and check every page.

R12-D. Before a new technical test table

LOCATION: Chapter III or IV > immediately above newly added test tables

FIND IN WORD: Insert beside relevant new SOL-SYS / extraction accuracy tables

ACTION: ADD introductory paragraph and a numbered table caption.

WHAT TO PUT INSTEAD:

Table [number] presents the selected test cases used to evaluate the corresponding system functionality. The table identifies the test-case reference, the expected behavior, the actual observed result, and the corresponding status. These results provide supporting technical evidence for the implemented features under the documented testing conditions.

RECOMMENDED REVISION ORDER

1. R1 system scope 2. R2 matching method 3. R3 approvals 4. R4 roster import 5. R5 status rules 6. R11 actual code 7. R6 testing 8. R8 extraction evaluation 9. R7 Objective No. 2 results 10. R9 conclusions 11. R10 finalized conceptual framework 12. R12 formatting.

SOURCE/VERIFICATION NOTE: This guide is based on the attached thesis DOCX and RevisionPaper.pdf (Dr. Solis targeted guidance). The original thesis is unchanged. The revision PDF does not cover every other panelist concern. Reported test counts are drawn from the revision guide rather than independently audited test artifacts; production status and real COR/ROG extraction accuracy require separate verification. Proposed DSRM framework and code-placement advice are additional guidance.